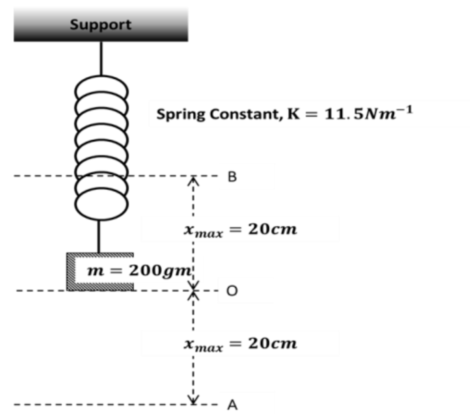
Khwaja Yunus Ali University

Spring-2021

Class Test – II (Physics-I)

Time: 45 Minutes **Part-A** **MCQ Marks: 15**

1. Investigate the following quantities from the spring pendulum shown in Figure.



a) Amplitude of Oscillation
b) Angular Frequency
c) Time Period
d) Frequency
e) Velocity at Point O
f) Acceleration at Point B
g) Total Energy
h) Where will the kinetic energy be highest?
i) Write down the differential wave equation for this pendulum.
j) Also write the solution of that differential equation.

2. Read the stem carefully and answer the questions that follows –

Two atoms ($m_1 = 2.6 \times 10^{-26}kg$ and $m_1 = 5.7 \times 10^{-26}kg$) are bound together with a force and they are oscillating about their equilibrium position. This oscillation can be replaced by a single particle SHM.

a) Find the reduced mass of the above system.
b) Justify the last statement of the stem

Time: 10 Minutes **Part-B** **MCQ Marks: 5**

1. What is the unit of the reduced mass?

a) $C^2N^{-1}m^{-2}$ b) Nm^2C^{-2} c) TmA^{-1} d) C e) kg

2. A force of 4N on the spring produces a displacement of 0.02m. What is the force constant of the spring?

a) $400Nm^{-1}$ b) $200Nm^{-1}$ c) $100Nm^{-1}$ d) $0.08Nm^{-1}$ e) $0.05Nm^{-1}$

3. Path difference produced by two wave is $\lambda/2$. what is the phase difference?

a) π b) $\pi/2$ c) $\pi/4$ d) $3\pi/2$ e) $3\pi/4$

4. Time period of the minute hand of the clock is –

a) $24\ hours$ b) $12\ hours$ c) $1\ hour$ d) $60\ min$ e) $60\ sec$

5. In case of SHO, Angular frequency, ω =?

a) K/m b) m/K c) $\sqrt{K/m}$ d) $\sqrt{m/K}$ e) K^2/m^2